

PIPPA MABY

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PROFILE

Third-year MEng Chemical Engineering student looking for a 2027 industrial placement, with a particular interest in sustainability and renewable energy. My EPQ on hydrogen as an alternative fuel sparked a genuine interest in research and problem-solving, which I have developed through university design projects and a recent fire-testing internship at Efectis. Over Summer 2026, I also taught myself Python and AI agent development despite having no prior programming experience, and developed an engineering agent to carry out process calculations using integrated tools. I enjoy applying my problem-solving skills alongside strong teamwork and communication skills.

EDUCATION

Loughborough University: MEng in Chemical Engineering **2024-2029**

- Year 2: 66% (2:1).

Ibstock Place School: London **2010-2024**

- **A-Levels:** Mathematics (A), Chemistry (B), Biology (C).
- **Extended Project Qualification (A*):** Is Hydrogen the Solution to the Climate Change Crisis?
- **11 GCSEs at grades 9-7, including 9** in Maths and 8s in all sciences and English Literature & Language.

ENGINEERING PROJECTS

Low-Alcohol Beer Process Design - group design project **2026**

- Designed a process to cut the alcohol content of beer below 0.5% while also recovering and purifying the extracted ethanol
- Modelled the process in Aspen Plus to determine optimal operating conditions for each unit operation; produced PFDs and P&IDs in Siemens NX.
- Delivered a full design report covering HAZOP, equipment datasheets, and economic and environmental analysis.

Propranolol Reactor Optimization - reaction engineering **2026**

- Modelled a 10 m³ batch reactor in MATLAB to investigate how operating strategy, temperature and reactant ratios affect conversion and selectivity
- Identified the operating conditions that maximized desired product and minimized by-products against an annual production target

Chemical Engineering AI Agent & Portfolio website - independent project **Summer 2026**

- Started with no programming background; self-taught Python, data science and AI agent development.
- Researched AI agent frameworks and tool calling, then built a chemical engineering agent capable of responding to queries and performing mathematical engineering calculations using integrated tools.
- Built and deployed personal portfolio website from scratch: <https://pippamaby.github.io/#projects>

WORK EXPERIENCE

Efectis, Cardiff - Intern **Sept 2026**

- Worked with fire testing engineers on the construction and preparation of fire resistance tests on specimens (walls and doors) to assess performance of materials and components.
- Gained an understanding of fire-testing equipment, procedures and standards, including the use of thermocouples for temperature measurement and the preparation of technical test reports.

HEA annual Hydrogen conference Volunteer **Jul 2026**

- Approached the organizers directly to secure a volunteer place and used the event to meet companies working across the hydrogen sector

High Society, London Waitressing at large high-end events, often 10-hour shifts **2025- Present**

Accommodation Committee Member: Loughborough University **2024- 2026**

GSK: Completed Molecule to Market module work experience **2023**

TECHNICAL SKILLS

- **Software:** Aspen Plus, Siemens NX 12.0, MATLAB, Microsoft Excel, Python
- **Engineering:** process design and optimization, HAZOP, PFD & P&ID development, economic and environmental analysis, laboratory experimentations and technical reporting

CERTIFICATES, ACHIEVEMENTS & VOLUNTEERING

- **Coursera (IBM), 2026:** AI Agents & Agentic Workflows, What is Data Science? AI Python for Beginners,
- **Awards:** Gold Duke of Edinburgh (2024) - Silver Industrial Cadet Award, renewable energy research (2023) - Jazz clarinet Grade 8, Merit (2023) - English Speaking Board Level 3 Grade 8, Distinction (2023)
- **Volunteering:** Full-time carer during post-operative recovery (Aug 2026) - Mosaic Primary School (2022-23) - Roehampton community Healthcare Program (2024).
- **Interests:** Tennis (Loughborough Halls team) - Fundraising (running for Cancer/Save the Children) - Cooking - Music.